

Using Gaussian Mixture Models to Classify Breast Cancer

Introduction

Breast cancer is the most commonly diagnosed cancer among women worldwide, responsible for an estimated 670,000 deaths globally in 2022.¹ In the United States, an estimated 321,910 new cases are projected for 2026, with approximately 42,140 deaths expected, representing 15.2% of all new cancer cases in the country.² Breast cancer is the second leading cause of cancer death in women in the United States and the chance that any woman will die from the disease is about 1 in 43, so improving classification tools is therefore one of the most important steps toward earlier detection and better outcomes.³ Breast cancer occurs when cells in the breast grow out of control, forming tumors that can spread to surrounding tissue and other parts of the body. Early and accurate classification of tumors as malignant or benign directly affects treatment decisions and patient outcomes.

Machine learning methods allow for a promising approach to tumor classification by identifying patterns in diagnostic measurements that may be difficult for clinicians to detect manually. In this project, Gaussian Mixture Models (GMM) were applied to the Wisconsin Breast Cancer Diagnostic Dataset to assess whether an unsupervised algorithm could identify the distinction between malignant and benign tumors from cellular measurements alone, without access to diagnosis labels during training.

Gaussian Mixture Models

A Gaussian Mixture Model (GMM) is an unsupervised machine learning algorithm that assumes data comes from a combination of K distinct groups, where each group is represented by its own Gaussian distribution. Rather than assigning each observation to a single cluster, GMM estimates the probability that each observation belongs to each group. The overall probability of observing a data point $\mathbf{x}$ is given by:

$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \quad k = 1, \dots, K$$

where π_k is the weight of cluster k and represents how much of the data that cluster accounts for and it must sum up to 1 across all clusters. $\boldsymbol{\mu}_k$ is the center of cluster k and $\boldsymbol{\Sigma}_k$ captures its shape and spread in feature space. Unlike K-Means, which forces every observation into exactly one cluster, GMM assigns each observation a probability of belonging to each cluster rather than a hard label. A point sitting between two clusters might be 70% one group and 30% another, reflecting the uncertainty rather than hiding it. This makes GMM especially well suited for biomedical data where classes naturally overlap. GMM assumes that observations are

independent, that data within each cluster follow a multivariate Gaussian distribution, and that the number of clusters K is specified prior to fitting.

Expectation-Maximization

GMM parameters are estimated via the Expectation-Maximization (EM) algorithm, which iterates between two steps until convergence. The E-step calculates the probability that each data point belongs to each cluster given the current Gaussian positions. The M-step then uses those probabilities to update each Gaussian's μ_k , Σ_k , and π_k to better explain the data. This process repeats until the model stops improving. While Expectation-Maximization is guaranteed to converge, it finds a local optimum rather than a global optimum depending on where it starts. To reduce this risk, the model was run 10 times with different random starting points and the best solution is kept.

Choosing K: Bayesian Information Criterion (BIC)

Since GMM requires K to be specified in advance, the Bayesian Information Criterion (BIC) was used to guide model selection. BIC rewards models that explain the data well but penalizes them for adding too many clusters and the lowest BIC score indicates the best balance. The GMM was fit for $K = 1$ through $K = 6$, resulting with the BIC values shown in Table 1.

Table 1. BIC scores for GMM with $K = 1$ through $K = 6$.

K	BIC
1	11,384.69
2	6,094.04
3	6,252.01
4	6,130.80
5	5,933.84
6	7,206.64

BIC dropped sharply from $K = 1$ to $K = 2$, which indicates that two dominant clusters are present in the data. Although the global BIC minimum occurred at $K = 5$, $K = 2$ was selected as the choice given the domain knowledge and that the clinical problem is inherently binary: tumors are classified as either malignant or benign. The $K = 5$ result suggests that sub-clusters may exist within each diagnostic class, possibly capturing differences within each diagnostic group that are beyond the scope of this analysis.

Dataset

The Wisconsin Breast Cancer Diagnostic Dataset was obtained from the UCI Machine Learning Repository and accessed via scikit-learn's built-in datasets module.⁴ The dataset contains 569 tumor samples, each described by 30 continuous features derived from images of fine needle aspirate samples. These features capture cell nuclear characteristics such as radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension. Each feature was measured three ways: as a mean, standard error, and worst value. This resulted in 30 features total and of the 569 samples, 212 are malignant (37.3%) and 357 are benign (62.7%). Before fitting the model, all 30 features were standardized to zero mean and unit variance to ensure no single feature dominated the Gaussian covariance estimates due to differences in scale.

Results and Discussion

A GMM with $K = 2$ was fit to the standardized features and each tumor was assigned to whichever cluster claimed it with the highest probability. Table 2 shows how those assignments compared to the true diagnosis labels.

Table 2. GMM Cluster Assignments vs. True Diagnosis Labels.

	Malignant (True)	Benign (True)
Cluster 0 (Malignant)	196	18
Cluster 1 (Benign)	16	339

The GMM with $K = 2$ achieved 94.0% accuracy, with only 34 of 569 samples assigned inconsistently with their true diagnosis. Cluster 0 captured 196 of 212 malignant tumors (92.5%) and Cluster 1 captured 339 of 357 benign tumors (95.0%). Precision and recall for the benign class were 95.0% and 94.96% respectively and the AUC was 0.9742, which indicates the model had a 97% chance of correctly distinguishing malignant from benign.

These results show that the Wisconsin Breast Cancer Diagnostic Dataset contains strong natural cluster structure that GMM was able to recover without ever seeing a diagnosis label. Although, several limitations should be noted. GMM assumes Gaussian cluster shapes which may not hold exactly for all 30 features and the BIC-optimal solution at $K = 5$ suggests the $K=2$ model may not capture the true structure in the data.

References

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